

ADEL MAGDY MACKNOTA

Full Stack Software Engineer | .NET & NestJS | Angular · React · Next.js

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PROFESSIONAL SUMMARY

Full Stack Software Engineer working across both the .NET and Node.js ecosystems. I build compliance-critical and financial systems end to end — ASP.NET Core and NestJS backends on SQL Server and PostgreSQL, with Angular, React and Next.js frontends. Current work includes a ZATCA e-invoicing integration hub, an automated payment reconciliation platform, and an AI-driven SEO automation engine for a ~750-product catalog — all three built to production standards and currently in active development ahead of release.

I care about architecture as much as working code: Clean Architecture, SOLID, idempotency, auditability, secure secret handling, and automated testing — so systems stay correct under real-world load and stay maintainable as they grow. I work fluently with AI development tools (Claude Code, ChatGPT, Gemini, Perplexity) to move faster on research, refactoring and testing, while keeping design decisions and code review under my own judgement.

TECHNICAL SKILLS

- **Languages:** C#, TypeScript, JavaScript, Python, SQL (T-SQL), HTML5, CSS3
- **Backend:** .NET 8 / ASP.NET Core Web API, NestJS, Node.js, Entity Framework Core, Prisma, LINQ, ADO.NET, REST APIs, SignalR, Hangfire, Redis/BullMQ, JWT & ASP.NET Core Identity, Dependency Injection
- **Frontend:** Angular 18, React, Next.js, TypeScript, RxJS, Bootstrap, Responsive Design, SPA
- **Databases & Caching:** Microsoft SQL Server, PostgreSQL, Redis
- **Architecture & Patterns:** Clean / Onion Architecture, Repository & Unit of Work, Specification Pattern, CQRS, MVC, SOLID, RBAC, Idempotency & Retry Design, Microservices (basic)
- **Integrations:** ZATCA E-Invoicing (Phase 2), Stripe, Microsoft Graph, Google Search Console API, LLM APIs (Gemini, Groq, Mistral, OpenRouter), Webhooks
- **DevOps & Tools:** Docker, Git & GitHub, CI/CD (basic), Postman, Visual Studio, VS Code, SSMS
- **AI-Assisted Development:** Claude Code, ChatGPT, Gemini, Perplexity — prompt engineering, AI-assisted refactoring, debugging, code review and documentation; used daily to shorten delivery cycles while keeping human review over every merged change
- **Testing & Methods:** Unit Testing, API & E2E Testing, Agile / Scrum, Code Review

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer

May 2026 – Present

Asus Card — Asas Al-Betaqat for Trading Est.

- Building three enterprise platforms end to end across two stacks (.NET/React and NestJS/Next.js) — owning architecture, database design, background processing, third-party integrations and automated testing. All three are in active development and not yet publicly live.
- Accelerated delivery by integrating AI tooling (Claude Code, ChatGPT, Gemini, Perplexity) into the workflow for specification research, scaffolding, refactoring, test generation and documentation — with every AI-assisted change reviewed and validated before merge.

ZATCA E-Invoicing Integration Hub — *In active development; validated against ZATCA Sandbox & Simulation*

Stack: NestJS, Next.js, TypeScript, PostgreSQL, Prisma, Redis/BullMQ, Docker

- Designed an integration hub that abstracts the full ZATCA e-invoicing compliance workflow behind one API, so client systems integrate once instead of implementing the specification themselves.
- Implemented the complete onboarding lifecycle: taxpayer/EGS registration, CSR generation and certificate handling, Compliance CSID, the 6/6 compliance checks, and Production CSID issuance.
- Delivered B2B Clearance and B2C Reporting flows with ICV/PIH sequencing, evidence tracking, and outbound webhooks for asynchronous status updates.
- Enforced strict environment isolation across Sandbox, Simulation and Production, with safeguards that block cross-environment and unauthorized ZATCA calls.
- Hardened the platform with RBAC, encrypted credential storage, idempotency keys, queue-based retries (Redis/BullMQ) and full audit logging.
- Covered the system with automated SDK, API and browser end-to-end tests, and packaged it for Docker-based deployment.

AI-Powered SEO Automation Platform — *In active development*

Stack: NestJS, Next.js, TypeScript, Prisma/PostgreSQL, Python, Google Search Console API, multi-provider LLMs

- Engineering an SEO intelligence and automation system for an e-commerce catalog of ~750 products, replacing fragmented manual optimization, stale SEO data and unsafe bulk updates.
- Designed a Family-first workflow that groups related products, detects SEO gaps with canonical gap tracking, and generates field-level AI recommendations.
- Implemented multi-provider LLM routing (Gemini with Groq, Mistral and OpenRouter fallback) plus a fact-validation and content-quality layer to guard against hallucinated output.
- Grounded recommendations in real performance data by integrating Google Search Console APIs, live product data, and Python-based scraping.
- Built controlled review → approval → apply pipelines with live verification and rollback, protected by idempotency, stale-data detection and permission checks.

Payment Reconciliation System — *In active development*
Stack: .NET, EF Core, React, SQL Server, Hangfire, SignalR, Microsoft Graph

- Designing and building an automated reconciliation platform to replace the manual comparison of bank settlements, payment-gateway data and merchant transactions.
- Automated ingestion of bank/HyperPay files via Microsoft Graph (email) and manual upload, with Hangfire background jobs backfilling missing Purple API data.
- Implemented a matching and exception-detection engine covering mismatches, missing transactions, refunds, duplicates, fees, commissions and net settlements.
- Delivered a real-time Live dashboard (SignalR) and a monthly Results view backed by atomic snapshot publishing for consistent, reproducible reporting.
- Prioritised financial accuracy, idempotency, auditability and secure secret handling, validated by automated end-to-end tests.

.NET Developer Intern Oct 2025 – Dec 2025
Smart Up — Cairo, Egypt (Hybrid)

- Contributed to the full SDLC of enterprise web applications using ASP.NET Core, handling backend logic and SQL Server database design across 3+ modules.
- Developed and optimized scalable RESTful APIs serving multiple frontend clients, reducing integration issues by ~40% through clean code and Agile standards.
- Collaborated with the frontend team to integrate REST APIs and resolve cross-stack issues through daily sync-ups and clear documentation.

PERSONAL PROJECTS

Store — E-Commerce RESTful API
Stack: .NET 8, ASP.NET Core Web API, EF Core, SQL Server, Redis, AutoMapper, Docker, Stripe · github.com/Macknota/E-Commerce

- Architected the backend with Clean (Onion) Architecture across 4 layers to enforce separation of concerns and SOLID principles.
- Implemented the Specification Pattern with a Generic Repository, reducing repeated query logic by ~60% across the codebase.
- Optimized performance with Redis caching for basket data, improving read response time by ~50% versus direct database queries.
- Secured 20+ endpoints with JWT authentication and role-based authorization; integrated Stripe payments and Docker deployment.

Social Connections Platform
Stack: .NET 8, Angular 18, SignalR, SQL Server, EF Core, ASP.NET Core Identity

- Built a responsive Angular 18 SPA with modular component architecture and RxJS, supporting 10+ reusable components.
- Implemented real-time messaging and live user-presence tracking with SignalR hubs, achieving sub-second message delivery.
- Designed a secure REST API (15+ endpoints) with JWT authentication, many-to-many likes/matches modelling, and cloud photo storage.

EDUCATION & TRAINING

B.Sc. in Computer Science — Benha University, Faculty of Science June 2023
Grade: Very Good · Military Service: Completed (Dec 2024)

Back-End Diploma — Route Academy 2025
Intensive training in C#, SQL Server, Entity Framework Core, and enterprise-level REST API development.

LANGUAGES

Arabic — Native | English — Professional Working Proficiency (B2) | Spanish — Beginner